01 / EDGE COMPUTE

Option CloudGate NXS

LTE variant | CN0124-12273 | Local processing, logging and helm UI

Gateway specifications

Processor	NXP i.MX 9332, dual Arm Cortex-A55 at 1.7 GHz. Optional neural processing unit.
Memory / storage	2 GB LPDDR4, 4 GB pSLC eMMC and internal microSD slot. Vendor-rated microSD ceiling is 2 TB and 104 MB/s.
Ethernet / serial	Two configurable 1 GbE RJ45 ports. Front serial expansion supports serial/Modbus. USB requires the Smart Metering expansion card.
Radio hardware	Telit LE910C4-WWXD LTE Cat 4, up to 150 Mb/s down and 50 Mb/s up. Two SMA ports for cellular and diversity/A-GNSS. Optional WLAN/BLE or LoRaWAN expansion.
Power input	9–50 V DC via 4-position Micro-Fit 3.0, with ignition/sense input. Optional last-gasp battery card supports up to one hour of limited functionality.
Trust hardware	NXP EdgeLock secure enclave and Infineon TPM 2.0. Verify the selected firmware and application boot path during integration.
Enclosure	Aluminium, 115 × 106 × 45 mm, 330 g. Operating range -25 to +75 °C. Bulkhead mounting or DIN rail with adapter.

Northwatch integration and bench acceptance

Receiver ingest	Use the P6 receiver as the GNSS source. For USB, validate gateway host mode, driver, power budget and recovery with the selected expansion card. For serial, match electrical levels, framing and throughput.
Private helm link	Serve the local read-only status UI over dedicated Ethernet. Verify the link and its power arrangements separately. Disable cellular and optional radios for the passive trial.
Logging runtime	Log raw receiver output, UTC and monotonic receive times, health and transitions. Measure log bytes/hour, divide usable storage by measured rate, and allow space for rotation. Measure total draw and endurance on the complete assembly.
Recovery checks	Test cold boot, receiver reconnect, stale input and storage pressure. Record time to first valid status, time to stale indication and logging coverage.

VENDOR REFERENCES

Option CloudGate NXS LTE datasheet

02 / GNSS FRONT END

Septentrio mosaic-G5 P6

Receiver module and mosaic-go G5 P6 evaluation kit

Receiver and kit specifications

Module / kit	LGA module 22.8 × 16.4 mm, 2.2 g. Evaluation kit 74 × 44 × 11.4 mm, 50 g, with USB-C, UART, PPS and SD slot.
Host interfaces	USB 2.0 High Speed device, two LVTTL UARTs up to 4 Mb/s, two configurable event inputs and one PPS output.
GNSS signals	GPS L1/L2/L5, Galileo E1/E5a/E5b/E6, GLONASS L1/L2/L3 CDMA, BeiDou B1/B2/B3, QZSS L1/L2/L5/L6, NavIC L5 and SBAS.
RF resilience	AIM+ Premium for interference monitoring and mitigation, with jamming/spoofing indications. Receiver features include RAIM+ and Galileo OSNMA support.
Position output	Up to 100 Hz positioning. Vendor horizontal/vertical RTK figures are 0.6 cm + 0.5 ppm and 1 cm + 1 ppm. Galileo HAS figures are 0.1 m horizontal and 0.2 m vertical under vendor conditions.
Antenna modes	Quad-band positioning in single-antenna mode. Dual-antenna mode provides triple-band positioning and heading plus pitch or roll. Vendor heading figures are 0.15° at 1 m and 0.03° at 5 m baseline.

Northwatch status and performance checks

Telemetry	Validate enabled receiver messages and record firmware and configuration. Capture fix type, position, data age, health and available interference indicators. Measure actual cadence and data volume at the selected output rate.
Alerts	Show suspected interference, fix loss and monitor unavailability as distinct states. Corroboration requires an independent fix. Record GNSS-reported and independently observed positions with source, time and uncertainty.
KPIs	Measure time to advisory, nuisance alerts per watch, fix availability, logging coverage and position disagreement where a timely reference exists. Record missing reference data explicitly.
Installation	Use a suitable external enclosure for deck exposure. Verify antenna band coverage, bias, connector type, RF loss, power budget, mounting and cable routing with the installer.

VENDOR REFERENCES

Septentrio mosaic-G5 P6 module specifications

Septentrio mosaic-go G5 P6 evaluation kit specifications